



Tennessee TECH

SAE Aero Design East 2026

Tennessee Technological University

Team 034: Golden Eagle 3

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Statement of Compliance

APPENDIX A - STATEMENT OF COMPLIANCE

Certification of Qualification

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Statement of Compliance

As faculty Adviser:

JT (Initial) I certify that the registered team members are enrolled in collegiate courses.

JT (Initial) I certify that this team has designed and constructed the radio-controlled aircraft in the past nine (9) months with the intention to use this aircraft in the 2026 SAE Aero Design competition, without direct assistance from professional engineers, R/C model experts, and/or related professionals.

JT (Initial) I certify that this year's Design Report has original content written by members of this year's team.

JT (Initial) I certify that all reused content has been properly referenced and is in compliance with the University's plagiarism and reuse policies.

JT (Initial) I certify that the team shall use the Requirements Check & Safety and Airworthiness Inspection checklists to inspect their aircraft before arrival at Technical Inspection and that the team shall submit the completed checklists, signed by the Faculty Advisor or Team Captain, to the inspectors before Technical Inspection begins.

John T. Tester
Signature of Faculty Advisor

1/12/2026
Date

Nicholas Ramsdell
Signature of Team Captain

1/12/2026
Date

Note: A copy of this statement needs to be included in your Design Report as page 2 (Reference Section 4.3)

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Table 1: Symbols and Acronyms

Symbol	Description
C_H	Horizontal tail volume coefficient
C_V	Vertical tail volume coefficient
S_H	Horizontal stabilizer planform (m^2)
S_V	Vertical stabilizer planform (m^2)
l_H	Distance from aircraft CG to horizontal tail aerodynamic center (m)
l_V	Distance from aircraft CG to vertical tail aerodynamic center (m)
c_{MAC}	Wing chord MAC (m)
b	Aircraft wingspan (m)
ρ	Total air density
p_d	Partial dry air pressure (Pa)
p_v	Partial water vapor pressure (Pa)
R_v	Specific gas constant for water vapor
R_d	Specific gas constant for dry air
Acronyms	Description
AoA	Angle of Attack
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
CG	Center of Gravity
DAQ	Data Acquisition
FEA	Finite Element Analysis
GE-2/ GE2	Golden Eagle 2
GE-3 / GE3	Golden Eagle 3
ISA	International Standard Atmosphere
MAC	Mean Aerodynamic Chord
NP	Neutral Point
ODE	Ordinary Differential Equation
PETG	Polyethylene Terephthalate Glycol
PPB	Payload Prediction Bonus
TLYF	Test Like You Fly
TPU	Thermoplastic Polyurethane

1.0 Executive Summary

This report details Tennessee Technological University's design decisions and accompanying research, testing, and modelling for the Golden Eagle 3 (GE 3); our submission into the 2026 SAE Aero Design East Regular Class competition. For this competition, the team must design a heavy lift remote-controlled aircraft capable of carrying unmodified 2L soda bottles. Scoring will be determined by the number of "empty" bottles weighing between 1 and 4 lbs and the number of "filled" bottles weighing more than 4 lbs; with additional bonus points available for accurately predicting flyable payload for a given density altitude. This year's team focused heavily on optimizing aircraft structure to decrease aircraft weight and increase available payload without going above the FAA aircraft limit of 55 lbs. This design focus is reflected in the aircraft's 18.42 lbs empty weight and a maximum flyable payload weight in the expected weather conditions in Lakeland of approximately 9.28 lbs, consisting of 2 "filled" bottles and 1 "empty" bottle.

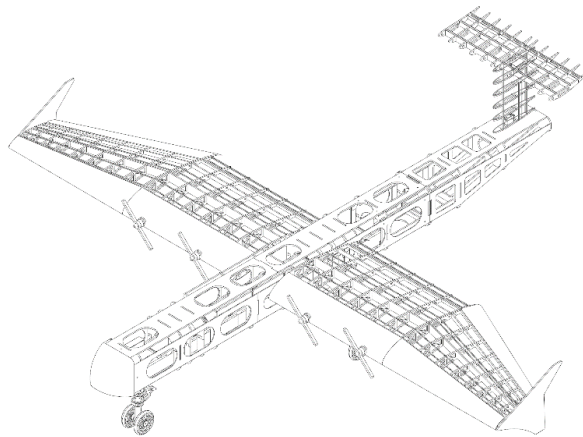


Figure 1: Isometric view of Golden Eagle 3

1.1 Competition Projections and Conclusions

Taking last year's GE-2's over built construction resulting in no flyable additional payload into consideration [1], a major design goal for the GE-3 was to optimize the structure to allocate for larger payload weight while not compromising structural integrity of the aircraft. Based on historical weather data shown in Section 5.2, the GE-3 is expected to take off with a maximum of 9.3 lbs of payload in 98.04 feet resulting in a single flight score of 34 points. Using the provided in Section 7.5 of the rulebook, our three best flights are tabulated below, resulting in a total flight score of 91.00 points before the payload prediction bonus. Our strategy for achieving this is detailed in Section 4.2.1.

Table 2: Top 3 expected single flight scores.

Number of "empty" bottles (1-4 lbs)	Number of "full" bottles (>4 lbs)	Total Weight (lbs)	Flight Score (No PPB)
3	1	7.37	27
0	2	8.19	30
1	2	9.28	34

1.2 Discriminators

To handle the increasing weight of the aircraft with incrementing payload, our front landing gear utilizes an internal spring to dampen shock loads travelling through the front landing gear and into the rest of the fuselage. While this requires more design time than previous landing gear configurations, the spring will mitigate the risk of damaging the fuselage on suboptimal front landing gear first landings.

To further improve takeoff performance, flaps have been installed into the inner 3 feet of the wing on either side of the fuselage allowing for more lift to be generated at slower speeds. These flaps extend 20% up the chord length from the trailing edge; similar to the ailerons located along the outer 3 feet of the wing trailing edge either side of the fuselage.

2.0 Schedule Summary

At the beginning of the 2026 competition season, the team held a 3-night session of classes dubbed, "Crash Course Airplanes" to establish a common vocabulary and baseline knowledge across the team, mitigating the risk of miscommunication error within design and manufacturing. Using the 2025

competition season schedule as a baseline [1], the 2026 design schedule was developed firstly as a backdating calendar with input from team leaders for realistic yet aggressive deadline expectations. Overall, deadlines were more aggressive compared to 2025 to allow ample time to review before the nonnegotiable competition deadlines. Furthermore, the Gantt chart shown in Figure 2 allowed for productivity to be optimized by organizing related tasks to finish simultaneously to reduce time wasted waiting to begin the next task. Where physically possible and in accordance to their impact on flight operations, a combination of simulations, and real-world “Test Like You Fly” tests were employed.

A month and a half into the design season, our first major material order was placed for a small number of materials to test and propulsion system candidates to allow for testing to begin during the latter half of the Fall 2025 college semester. This came with the added benefit of an opportunity to become familiar with the manufacturing methods to improve manufacturability and gauge the quality of this year’s available stock.

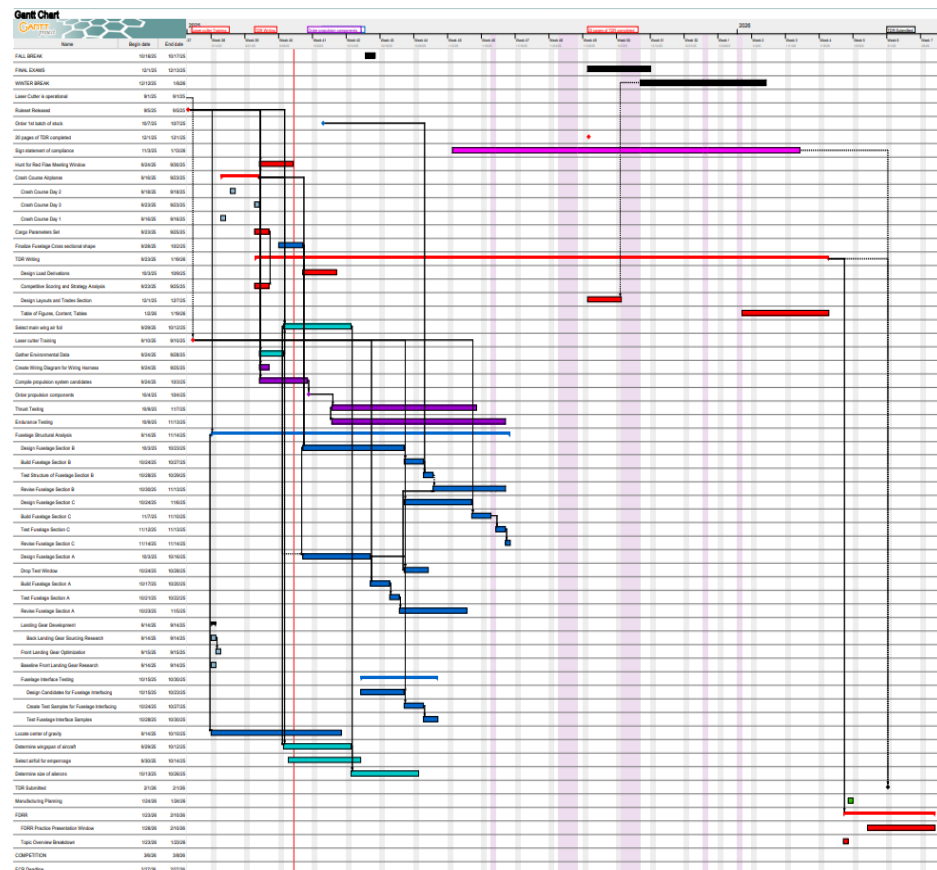


Figure 2: Design Season Gantt Chart

3.0 Table of Referenced Documents, References, and Specifications

Table 3: Table of References

Reference	Description/Purpose
[1] K. Albrecht, A. Savage, et al., “SAE Aero DesignTennessee Technological Univeristy Technical Design Report,” Jan. 2025.	Referencing previous work from the club.
[2] US, “Time Series Viewer,” <i>Weather.gov</i> , 2026. https://www.weather.gov/wrh/timeseries?site=KLAL (accessed Jan. 30, 2026).	Gather weather data around competition site
[3] P. Barua, T. Sousa, and D. Scholz, “Empennage Statistics and Sizing Methods for Dorsal Fins,” Hochschule fur Angewandte Wissenschaften Hamburg, Apr. 2013. Accessed: Jan. 31, 2026. [Online]. Available: https://www.fzt.haw-hamburg.de/pers/Scholz/Aero/AERO_TN_TailSizing_13-04-15.pdf	Equations for empennage sizing coefficients
[4] SAE Aero Design, “2026 SAE Aero Design Rules,” Sep. 2025. https://saeaerodesign.com/cdsweb/gen/DocumentResources.aspx (accessed Oct. 01, 2025).	Competition rules and requirements
Anderson, J.D (2011) <i>Fundamentals of aerodynamics</i> (5 th ed.). McGraw-Hill	Aerodynamic principles and fundamentals
Sadraey, M. H (2013) <i>Aircraft design: A systems engineering approach</i> (1 st ed.) Wiley.	Fundamentals of aircraft design and system integration

4.0 Design Layout and Trades

Upon reviewing Tennessee Tech’s 2025 competition aircraft, the Golden Eagle 2, a major driving factor for design decisions outside the 2026 Regular class mission is to optimize the aircraft structure to maintain the structural stability of the aircraft while reducing unnecessary weight. While the competition scoring no longer favors larger wingspan, the goal remains to be accomplished when comparing the 34 lbs empty weight of the GE-2 to the 18.8 lbs empty weight of the GE-3 [1].

The team opted for a mix of 3D printed and laser cut subassemblies to reduce the design limitations stemming from manufacturability and focus primarily on performance. Such examples of this decision include the 3D printed nose cone to help shift the CG forward of the neutral point (NP) and the 3D printed ailerons and flaps on the main wing.

4.1 Overall Design Layout and Size

The Golden Eagle 3 has a total wingspan of 119 inches with a trapezoidal planform, allowing for 0.5-inch margin on either side of the wing to account for unforeseen manufacturing errors. From nose to tail, the aircraft is 107.5 inches, allowing for ample room for the payload bay running most of the fuselage length. Empty CG for the craft is located 40.30 inches back from the nose of the craft and 4.5 inches in front of the neutral point of the wing. At the aircraft's highest point, it stands 31.5 inches tall and is predominantly constructed from balsa and aero-lite plywood with specific parts being 3D printed or made from aluminum.

4.2 Optimization

4.2.1 Competitive Scoring and Strategy Analysis

To satisfy the payload requirements outlined in section 7.4 of the rulebook, “empty” payload must weigh between 1 lbs and 4 lbs with anything above 4lbs being considered “full”. Satisfying this rule without carrying unnecessary weight requires sourcing a material with a theoretical density of 929 g/L as this would fill a 2L bottle and weigh 4.1 lbs. Alternatively, the team chose to use pea gravel with a density of 1201 g/L to fill each bottle 0.1 lbs heavier than the minimum weight required to satisfy being “empty” or “full”. Reducing the risk of pea gravel shifting during flight, cotton balls will be packed into each bottle to fill the gaps in the bottle.

4.3 Design Features and Details

Our team chose the S1223 airfoil design for the wing due to its high lift properties at low Reynolds numbers. The ailerons are 20% the length of the chord. For the horizontal stabilizer, we chose the inverted NACA 2412 airfoil to balance the moment generated by the main wing in level flight. The elevator on the horizontal stabilizer is 30% of the chord length. We chose a T-tail design to place the empennage above the disturbed airflow trailing from the main wing. This design works well at lower to moderate angles of attack. However, higher angles of attack (AoA) increase the risk of the wing blocking airflow to the horizontal stabilizer resulting in degraded performance. For this competition however, we do not expect to operate at higher AoA. To reduce drag from wing tip vortices and balance manufacturability, we chose

to implement wingtip fences made from 0.25” balsa. The wing structure is primarily composed of balsa wood, chosen for its low density compared to other wood. As shown in Figure 3, to maintain strength and mitigate the risk of the rib structure breaking at the narrow trailing edge of the S1223 airfoil, PETG will be used to withstand the aerodynamic forces during flap deflection, as it has a higher tolerance than wood under smaller cross-section.

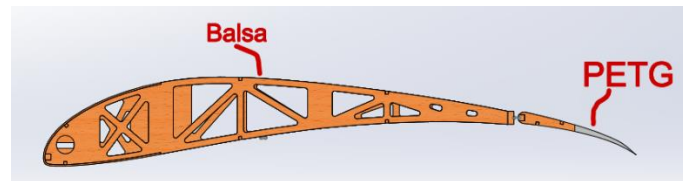


Figure 3: Cross sectional view of wing emphasizing transition from Balsa to PETG along airfoil.

Balsa is used for most of the side walls of the fuselage for its ability to withstand the expected pressure. For aircraft sections expected to experience higher pressure such as structural sub system connection points as well as the fuselage ceiling and floor, plywood will be installed for its higher durability and strength relative to Balsa.

4.4 Interfaces and Attachments

With the design constraint of transporting the aircraft to Lakeland Florida, the team investigated implementing wood joinery to be able to deconstruct the aircraft for travelling to and from the airfield. Secondly, the competition requirement to unload payload in under 1 minute required the fuselage to be predominately empty and unobstructed space for the payload bay. Having 2 people available to fully unload the aircraft, the payload will be removed through the nose of the aircraft, with the nose being fastened to the fuselage via nylon bolts capable of withstanding 70 lbf of shear force.

To accommodate the 8”x8” cross sectional fuselage area necessary for the payload bay and structure in the fuselage using the commercially available Balsa dimensions, the team elected to add dovetail joints to connect the top and bottom bulkhead halves. FEA determined the optimal dovetail joint angle to be 21 degrees however due to a miscommunication from design to manufacturing, the dovetail

joints are angled to 19 degrees. This difference in angle however has negligible impact on performance as found through stress analysis.

5.0 Loads and Environments

5.1 Design Load Derivations

Evaluating the structural integrity of the plane, the plane was modeled as a hollow beam with a square cross-section ($W \times H$) and a uniform wall thickness (t). The critical loading scenarios, Steady State Flight and Static Ground Loading, were analyzed using static equilibrium equations.

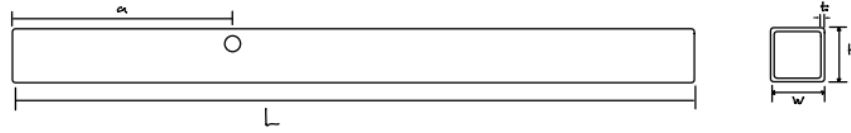


Figure 4: Simplified Model of Plane with Spar Location (a) Marked

The fuselage weight was split into three sections (S_A , S_B , S_C) and the payload (P), all modeled as uniformly distributed loads. Components such as the wing, empennage, and landing gear were treated as point loads acting at their respective locations (x).

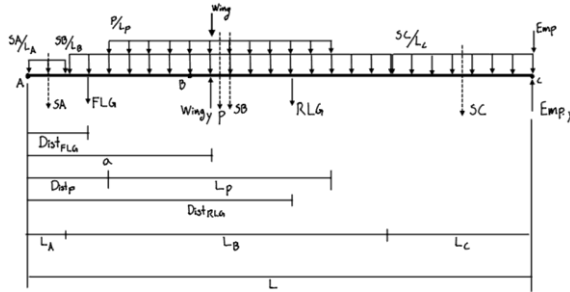


Figure 5: Free Body Diagram of In-Flight Model

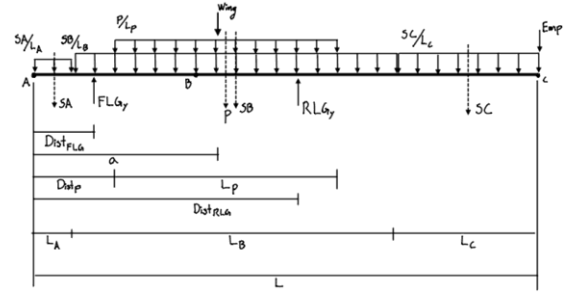


Figure 6: Free Body Diagram of On-Ground Model

The analysis utilized a payload configuration of one full and two empty bottles, as that is the estimated maximum payload the plane can safely carry. The In-Flight Model shown in Appendix A determines the required lift forces during steady-state flight. The wings and empennage act as the primary supports for the plane and payload weight. From this, it was determined lift needed from the

wings to support the payload is 22.742 lbf, and the empennage must produce a downward force of 0.867 lbf. The On-Ground Model shown above in Appendix A evaluates the reaction forces of the landing gear. The front (*FLG*) and rear (*RLG*) landing gear act as support for the stationary plane. From this, front and rear landing gear must be able to withstand 4.866 lbf and 14.624 lbf, respectively, when stationary.

Shear force and bending moment diagrams were generated for both load distribution scenarios. The analysis revealed that the maximum bending moment consistently occurs at the wing spar. To account for this, the area around the spar was reinforced with $\frac{1}{4}$ inch plywood inserts. Shear and bending moment diagrams can be seen below with $x=0$ inches and $x=120$ inches representing the nose and tail respectively.

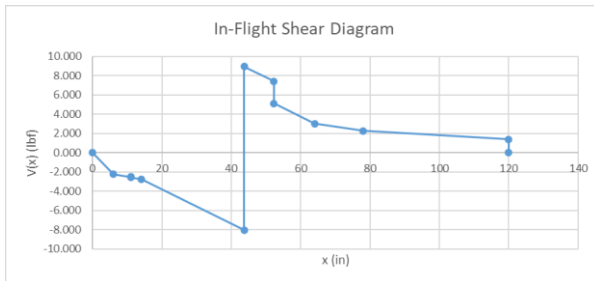


Figure A: In Flight Shear Diagram for fuselage

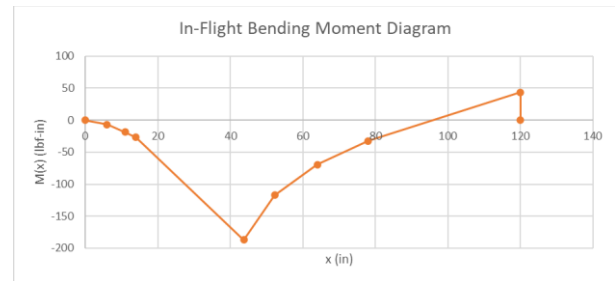


Figure B: In-flight bending moment diagram for fuselage

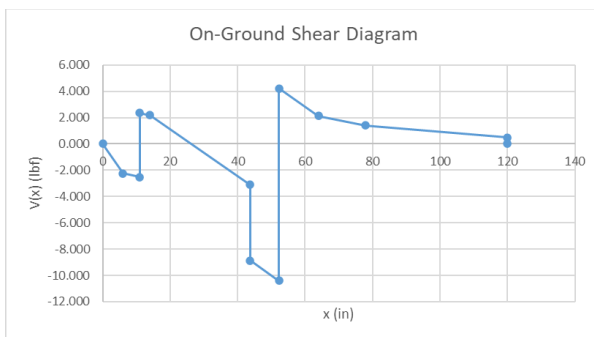


Figure C: On-Ground shear diagram for fuselage

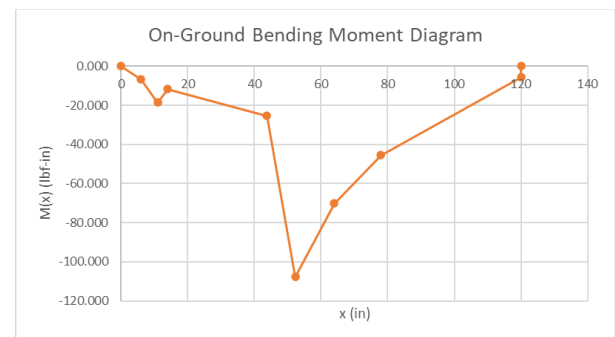


Figure D: On-ground bending moment diagram for fuselage

Figure 7: In flight moment and shear diagram (Figure A and B) and on ground shear/moment diagrams (C and D)

5.2 Environmental Considerations

Our team is participating in SAE Aero Design East 2026 in Lakeland, Florida in early March. Thus, the team researched historical weather conditions for February 29-March 10 in 2021-2025 from Lakeland Linder International Airport [2]. Thus, total average temperature, dew point, and percent relative humidity were calculated from each year. From there, total average air density was calculated using Eq. (5.2a) to help us estimate what our expected cruising velocity is. Air density values were then used to find the cruising velocities in different conditions. Equations 5.2a and 5.2b shown below were applied to plot expected cruising velocity as a function of temperature at varying humidity levels as plotted in Figure 7.

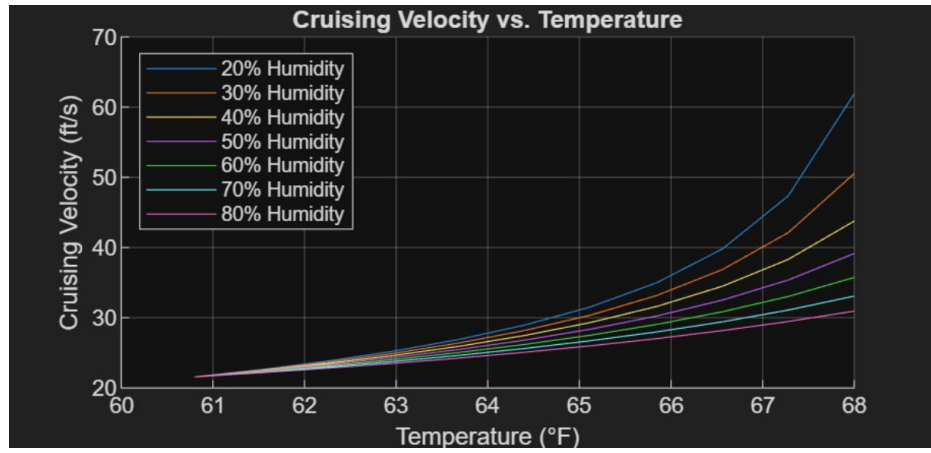


Figure 8: Cruising Velocity vs. Temperature at various humidities

$$\rho = \frac{p_d}{R_d T} + \frac{p_v}{R_v T} \quad (\text{Equation 5.2a})$$

$$v = \sqrt{\frac{2W}{\rho C_L A}} \quad (\text{Equation 5.2b})$$

The historical data suggests that the air density for the competition should be around 0.00238 slug/ft³ with an average temperature of 65°F. Additionally, air density directly affects dynamic thrust. Thrust is proportional to the mass of the air. As air density increases, the propulsion reaction from the propeller increases, leading to improved dynamic thrust performance.

6.0 Analysis

6.1 Analysis Techniques

Throughout the season, design tools and scripts were developed to decrease turnaround times between design revisions. With this, the team moved away from ‘rule of thumb’ reasoning without numerical validation to create a clearer benchmark process to catalog trends for improvements. When feasible within the timeline or required, the team elected to use Test Like You Fly (TLYF) testing procedures to validate design decisions and models. Applying TLYF allowed sub systems to be developed despite lacking knowledge unobtainable within the timeline.

6.1.1 Analysis Tools

For overall structural CAD modelling and FEA, the team elected to use SOLIDWORKS once general dimensions and parameters were determined through Excel design tools and MATLAB scripts. Preliminary wing development was done using XFLR5 software before moving into Ansys Fluent for higher resolution and accuracy and validating the location of the empennage relative to the wake of the main wing.

When simulations became more resource intensive than testing, as was the case for determining rolling friction and static thrust, the team opted for real world testing utilizing TLYF principles of testing at the lowest possible system first before moving to more integrated systems.

6.1.2 Developed Models

Continuing to improve our analytical tools and techniques, the team put considerable time into developing mathematical models and simulations to predict aircraft performance prior to manufacturing and first flight. Such models include modelling runway performance in section 6.2.1 and utilizing FEA analysis for dovetail joints. With this however, there is inherent risk as these models lack validation and cannot model unforeseen circumstances with low probability of occurrence, the team is over all confident in the accuracy of the models and will continue to develop, improve, and validate these models to enhance the team’s design tools and techniques.

6.2 Performance Analysis

6.2.1 Runway Performance

To evaluate the GE3's runway performance, the team performed a series of CFD simulations to determine the lift and drag characteristics of the final configuration. These values were then combined with the measured rolling-friction coefficient of the landing gear to develop a takeoff velocity model. Using equation (6.2.1a) below, the resulting forces were combined and plotted in MATLAB to show takeoff distance as a function of payload.

$$\Sigma F_x = T(v) - D(v) - f(v) \quad (\text{Equation 6.2.1a})$$
$$\Rightarrow v' = \frac{T(v) - F(v) - f(v)}{m}$$

The analysis assumes a no-wind environment and an air density of 0.0023 slug/ft³, consistent with the environmental considerations outlined in Section 5.2. Ground effect was not explicitly modeled, which may slightly reduce the actual takeoff distance. The rolling-friction coefficient, initially calculated at 0.05, was conservatively increased to 0.1 to incorporate a factor of safety. Solving the velocity ODE above we were able to compute the lift generated along the runway and identify the distance at which the aircraft achieves sufficient lift for takeoff during each payload configuration. Takeoff was defined as the point where $L = W$, at which the normal force and rolling friction reduced to zero. Our findings were plotted in Figure 9.

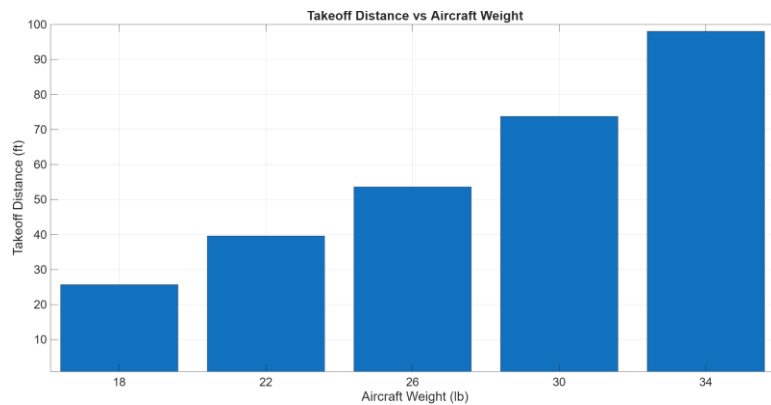


Figure 9: Takeoff distance (ft) vs Aircraft Weight (lbs)

Landing performance was evaluated using the same equation applied in the takeoff analysis. By setting the initial velocity equal to the cruising speed, estimated to be 23 ft/s, and applying $T(v) = 0$, the model provided an estimate of the required landing distance for each payload configuration. These findings were plotted in Figure 10.

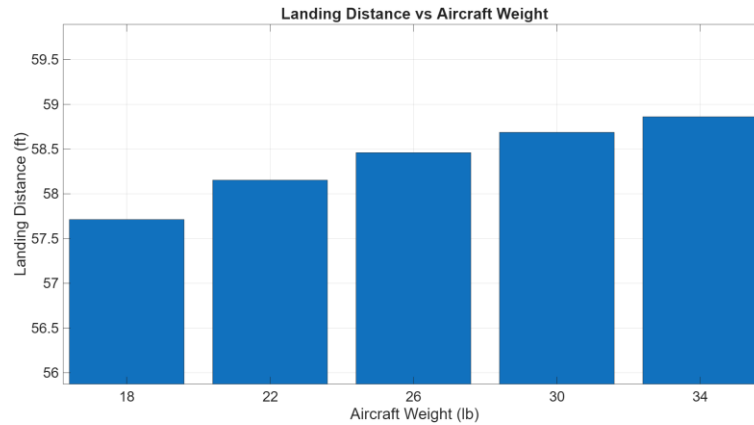


Figure 10: Landing distance (ft) vs. aircraft weight (lbs)

6.2.2 Flight and Maneuver Performance

A T-tail empennage configuration was selected to improve pitch control authority by placing the horizontal stabilizer above the wing wake and fuselage boundary layer. Allowing the stabilizer to operate in undisturbed airflow during normal flight conditions, resulting in consistent elevator response. Although T-tail configurations may experience reduced effectiveness at higher angles of attack, the GE3 is expected to operate below these conditions.

The horizontal stabilizer uses an inverted NACA 5412 airfoil, selected to balance the pitching moment of the aircraft in stable flight. The vertical stabilizer utilizes a NACA 0010 airfoil, providing neutral yawing moment characteristics, predictable rudder response, and less drag than the previously considered NACA 0012 vertical stabilizers. Empennage sizing was performed using tail volume coefficient equations to ensure adequate longitudinal and directional stability while minimizing added

weight and drag. Stabilizer area was found by plugging in favorable coefficients and lever arm distances that met our criteria to the equations below [3].

$$C_H = \frac{S_H l_H}{S_W c_{MAC}} \quad (\text{Equation 6.2.2a})$$

$$C_V = \frac{S_V l_V}{S_W b} \quad (\text{Equation 6.2.2b})$$

Maneuverability targets were established to ensure responsive but controllable handling. The aircraft is designed to achieve a roll rate of 45 deg/s with 90 deg/s² acceleration, a pitch rate of 30 deg/s with 60 deg/s² acceleration, and a yaw rate of 15 deg/s with 30 deg/s² acceleration.

Performance analysis indicates stable flight at a cruising velocity of 19.6 ft/s, with sufficient thrust for takeoff and climb. Predicted takeoff distance and climb rate meet competition requirements. GE3 is expected to demonstrate reliable maneuverability and stable performance throughout its flight.

6.2.3 Shading/Downwash

Ansys Fluent was used to analyze the flow characteristics around a 2D wing and horizontal stabilizer airfoil positioned at the geometric locations of the design of the aircraft to determine the possibility of adverse flow interaction between the two components. Such interaction of the main wing slipstream on the empennage may cause premature stall or unpredictable flow around the stabilizer and control surface, contributing to a partial or total loss of control. As the horizontal stabilizer has been placed at a larger distance away from the wing as compared to previous years, the effect of the slipstream on this component is no longer a concern. A velocity profile of the simulation also backs this claim, as the large area between the slipstream and stabilizer are adequately distanced as shown in Figure 11.

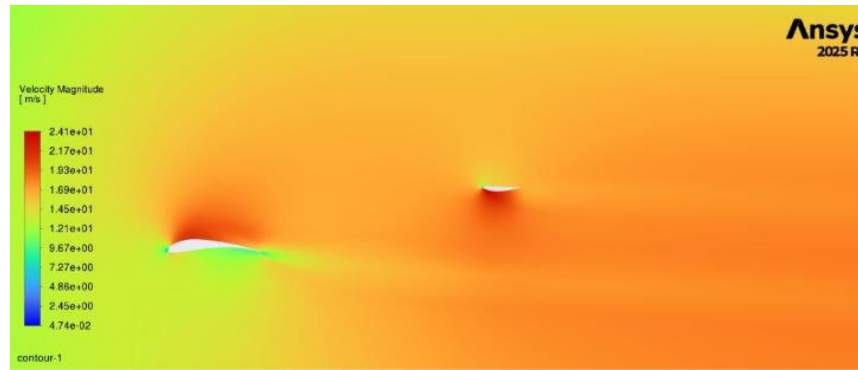


Figure 11: Flow characteristics of main wing and horizontal stabilizer.

6.2.4 Dynamic and Static Stability

The GE3 ensures positive longitudinal static stability by placing the CG 3.2" ahead of the neutral point (NP) of the wing. The wing NP is 6" back from the leading edge. Due to uncertainty in calculating the net NP of the GE3, we opted to take the NP of the wing as the basis for the static margin calculation to guarantee that the GE3 will be positively stable regardless of the accuracy of the hand calculation. Equation 6.2.4a relates the NP and CG to give a static margin of .0214, which will sufficiently aid in stall recovery.

$$SM = \frac{x_{NP} - x_{CG}}{MAC} = \frac{6'' - 1.5''}{21''} = .214 \quad (\text{Equation 6.2.4a})$$

Proper control surface sizing and careful construction will result in a symmetric craft that is stable in the roll axis. The weathercocking effect produced by the rudder and fuselage provides stability in yaw. The use of gyro provides a safety net in case of unanticipated instability along any axis.

6.2.5 Aeroelasticity

A major design goal for the GE3 was to reduce weight; however, the design process must also consider the structural rigidity of the wing and its reaction to aeroelastic forces. Aeroelastic forces result from the combination of aerodynamic forces on the wing, with elastic and inertial forces from the reaction of the internal structure. Under certain circumstances, the forces imparted on the structure can grow exponentially to the point of structural failure.

Finite element analysis (FEA) was used to determine the rigidity of the wing structure and deflection under load. Lifting force loads on each rib were calculated by sectioning off each area that the rib is centered on and solving the lift force under each rib using the lifting force equation. Analysis concluded that at cruising velocity of 19.6 ft/s the maximum deflection of the wing under ideal circumstances is 0.08 inches, occurring at the trailing edge of the wing tips. The lifting force is increased by a factor of 3 to analyze edge cases concerning higher velocity or other spikes in force. Figures 12 and 13 show the displacement under this increased force, showing a maximum displacement of 0.25 inches.

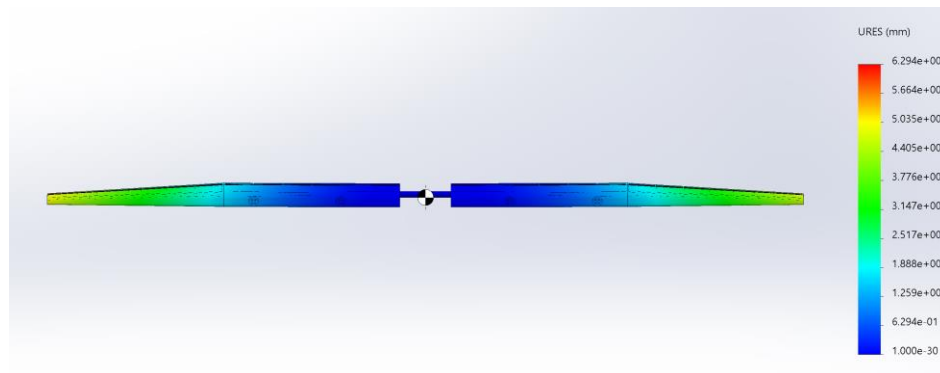


Figure 12: Aeroelastic displacement front view

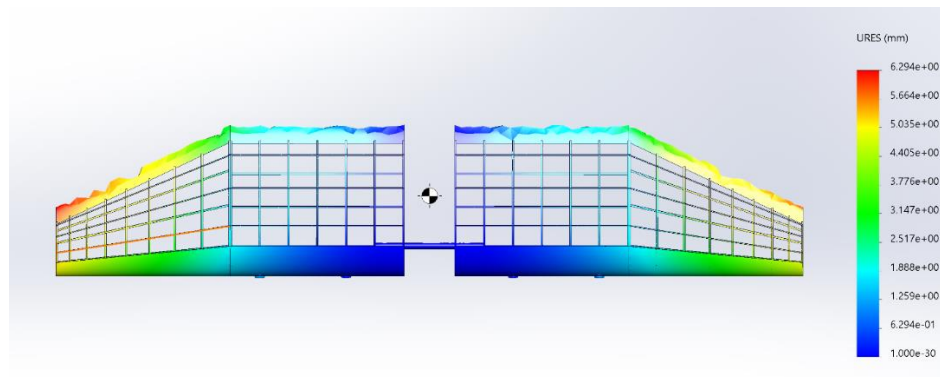


Figure 13: Aeroelastic displacement top view

Given this small displacement of the wing structure, elastic reactions will be insignificant compared to the lifting force. Aeroelastic forces will be minimized as a result, and the team predicts the risk of divergence is negligible.

6.2.6 Lifting Performance, Payload Predictions, and Margin

We analyzed historical weather conditions in Lakeland, Florida for February 29-March 10 across the years 2021-2025 using data from Lakeland Linder International Airport [2]. These records indicate temperatures between 50- 70 °F and an expected air density of approximately 0.00238 slug/ft³. Using the takeoff performance method described in Section 6.2.1, we computed the takeoff distance for each payload configuration. Operating at sea-level ISA conditions (59F, 14.7 psi), the predicted maximum payload is approximately 9.28 lbs taking off in 61 feet, corresponding to 2 full bottles and 1 “empty” bottle. Should the weather conditions improve compared to previous years, the maximum payload will trend towards the absolute maximum of 4 full bottles while still taking off within 100ft. Allotting for a 20 ft margin, and assuming optimum flight conditions based on previous years' weather data, the aircraft is predicted to be capable of taking off in 80 ft carrying 6 lbs of payload, equating to 1 full bottle and 2 "empty" bottles.

6.3 Structural Analysis

At its thinnest, the Golden Eagle 2 fuselage airframe is 0.125 inches thick balsa wood. Balsa wood was selected for much of the aircraft for its low density and ability to withstand distributed load before failure. In areas of the aircraft such as the wing and fuselage connection, 0.25” plywood is added for increased support at higher concentrated loads. Along with plywood sheets, bulkhead density is also increased at points expected to experience more concentrated loads, such as fuselage section joints and interfacing points between landing gear, fuselage, and wing.

6.3.1 Applied Loads and Critical Margins

As predicted by the team and shown in section 5.1, the largest loads acting on the fuselage will be concentrated at the wing spar. To accommodate this, a 0.125-inch-thick sheet of plywood will be installed at the wing spar attachment point along the fuselage reinforcing the already existing 0.125” balsa.

6.3.2 Mass Properties and Balance

The center of gravity for the GE-3 is located 41.61 inches down the longitudinal axis of the aircraft from the datum reference at the nose and 3.2 inches in front of the neutral point of the wing. To ensure stability for ground operations with smaller turning radius, the landing gear is arranged in a tricycle configuration where the empty weight CG is at a minimum 8.9 inches from the edge of the landing gear's triangular footprint.

To simplify unloading under time restrictions laid out in rules section 7.5 [3], payload will be loaded and unloaded through the nose via a rail and cart system. By installing blocks at carefully determined points along the rail, the payload sub system CG can be located directly above the aircraft empty weight CG location minimizing adverse effects on the CG from increasing payload weight.

7.0 Assembly and Subassembly

7.1 Electric Propulsion Summary

7.1.1 Maximum Static Thrust

To determine the propulsion system for the GE3, the electrical sub team designed a propulsion test stand to collect data on static thrust and power efficiency for each motor listed in **Table 4**.

Table 4: Efficiency ratios and Maximum Thrust and Current results for tested Motor and Propeller combinations

Motor / Voltage	Prop	Max Static Thrust (lbs)	Max Current Draw (A)	lbf/A
3110 / 1150 kV	9x9	2.38	50.48	0.047
3106 / 1500 kV	9x3	3.09	36.62	0.084
3106 /1500 kV	9x6e	3.60	52.35	0.069
3106 /1200 kV	9x6e	3.57	39.68	0.090

2320/1100 kV	9x4.5e	3.82	37.55	0.102
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The motor/prop combination we chose for our craft was the SkyWalker 2320/1100 kV motor with the 9x4.5e prop. This combination has the highest lbf/A ratio measured at 0.102 lbf/A as shown in Table 4, giving the greatest advantage at takeoff. Considering all four motors, full open throttle is expected to produce 15.28 lbf for takeoff. Extrapolating the current draw to 4 motors running simultaneously, we estimate the total amount of current draw for all four motors to be 150.2 A. However, the high continuous current poses a risk to the performance and health of the battery, which is considered in Section 7.1.3.

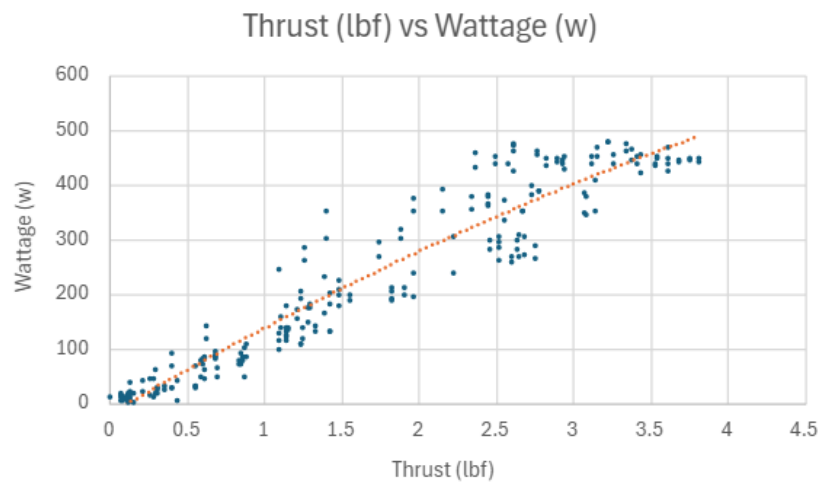


Figure 14: Thrust (lbf) vs Wattage (W) for 2320/1100 Kv motor with 9x4.5E propeller.

Figure 14 is a representation of our static max thrust values collected from the SkyWalker 2320/1100 kV motor with the 9x4.5e prop. The max static thrust for a single motor is 3.82 lbs at 450.68 watts.

7.1.2 Dynamic Thrust Predictions

Figure 15 represents the estimated dynamic thrust of a single motor. An estimated 3.50 lbs of dynamic thrust is expected at our takeoff distance of 30 ft/s for a single motor. This equation assumes ideal conditions, resulting in a 15-30% error when compared to actual dynamic thrust performance.

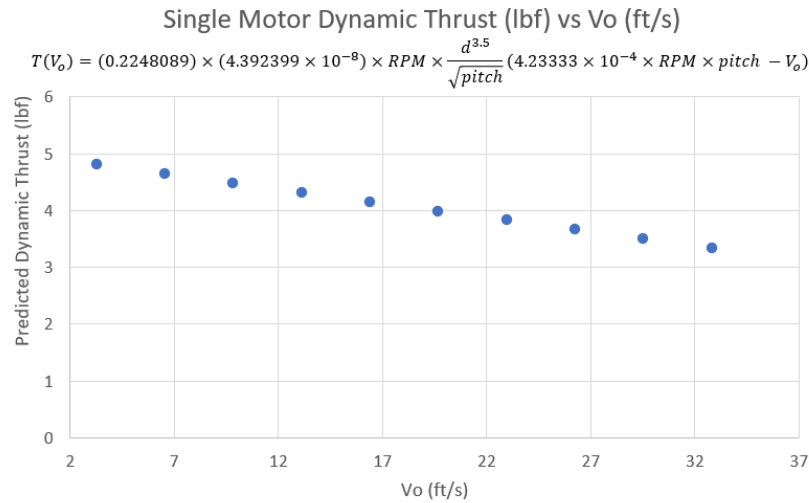


Figure 15: Single motor dynamic thrust predictions in 2ft/s to 32 ft/s air speed

7.1.3 Maximum Static Endurance

Endurance testing was conducted after selecting a suitable propulsion system candidate through maximum thrust testing outlined in section 7.1.1. The Hobbywing Skywalker 2320 1100kv paired with an APC 9x4.5E propellor was selected for its higher efficiency and lower current draw relative to the other configurations tested and tabulated in Table 4. All tests were conducted with a Liperior 2200mAh 4s battery charged to 4.20 volts per cell. The same battery was used to minimize variability in the tests. During each test, throttle values, thrust, battery current, and battery voltage were recorded at 2 Hz. Figure 16 illustrates total thrust output compared to time with 4 Hobbywing Skywalker 2320 1100kv motors over the duration of the endurance test.

Table 5: Endurance profile for 2320 1100 Kv motor and 4 9x4.5E propellers

Motor Prop	Average Thrust (lbs)	Max Thrust (lbs)	Max Current Draw (A)	Runtime (s)
Hobbywing 2320 1100kV APC9x4.5E	8.46	10.16	64.4	112.8

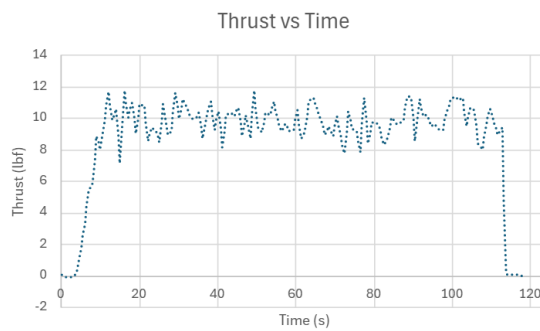


Figure 16: Thrust (lbf) vs Time (s) graph for a static endurance test.

The data follows the trends observed during maximum thrust testing. The test produced approximately 8.46 lbf at 64.4 A, consistent with expected performance for the motor-propellor combination.

8.0 Testing and Integration

8.1 Static Thrust Testing Stand

To determine static thrust values presented in Section 7.1.1, the electrical sub team elected to continue in-house development of a static thrust test stand. While this proved required more financial investment into the sub team, developing the test stand reduced uncertainty and time spent validating mathematical models with the added benefit of keeping electrical team tasks manageable within the competition design timeline.

The static thrust test stand consists of a mechanical arm with 1:1 advantage to redirect horizontal forces produced by the motor into vertical forces directed into a strain gauge as shown in Appendix A. Controls and DAQ are managed by an Arduino Nano to reduce human error in collecting current draw, voltage, and thrust data. As necessary, manual switches are hardwired into the circuit for manual override in ending a test prematurely. A circuit diagram of the test stand is shown in Appendix A.

8.2 Material Property Testing

Manufacturing the aircraft, the team decided to use up leftover stock since we already had the material and there has been no known exposure of the material to moisture. Furthermore, the team

became accustomed to using the 2025 competition material profiles for SOLIDWORKS modelling. The materials we used for the aircraft are balsa and plywood. Balsa was used for its lightweight maneuverability, and plywood was used in the sections we needed more structural stability. This analysis focuses on the volume, mass, weight, and density of balsa and plywood.

Table 6: Density values for various dimensions of balsa and plywood

Density					
Size	Material	Volume (m ³)	Mass (g)	Weight (N)	Density (kg/m ³)
1/8"x12"x48"	Ply	0.00117987	479.2	4.700952	406.14644
1/4"x12"x48"	Ply	0.00235974	999.33333	9.8034599673	423.49298
1/4"x8"x36"	Balsa	0.00117987	169.33333	1.6611599673	143.51863
1/4"x6"x36"	Balsa	0.0008849	129.8125	1.273460625	146.69737

Table 6 above shows each material and their chosen sizes, with calculations being done in the metric system for overall consistency. The plywood is, more specifically, 1/4" thick aero-lite birch plywood, which is used for the areas undergoing the most stress, with a density of roughly 423 kg/m³, and a weight of 9.8 Newtons, giving us the stability required. We used 1/4" thick balsa for sections experiencing less stress, with a calculated density of about 143-146 kg/m³, and a weight between 1.27-1.66 Newtons.

9.0 Conclusion

Within the competition scope and looking ahead towards March 6-9th, the primary goal for the Golden Eagle 3 is to carry our predicted maximum payload of 9lbs. As a team in their third year of existence, we have also made extensive effort and progress in documenting and overall improving the design process. As stated throughout the report under the operating conditions expected at Lakeland, Florida, the GE-3 projects to carry approximately 9 lbs and generate 15.28lbf of static thrust at takeoff before taking off in approximately 60ft before cruising at 19.6 ft/s for an expected runtime of 112 seconds. All culminating in a projected total flight score of 91.00.

Appendix A

Calculations:

$$\uparrow \sum M_O = 0$$

$$SA(a - \frac{L_A}{2}) + FLG(a - Dist_{FLG}) - P(Dist_P - \frac{L_P}{2} - a) - SB(L_A + \frac{L_B}{2} - a) - RLG(Dist_{RLG} - a) - SC(L_A + L_B + \frac{L_C}{2} - a) - Emp(L - a) - Emp_y(L - a) = 0$$

$$(L - a)Emp_y = -SA(a - \frac{L_A}{2}) - FLG(a - Dist_{FLG}) + P(Dist_P - \frac{L_P}{2} - a) + SB(L_A + \frac{L_B}{2} - a) + RLG(Dist_{RLG} - a) + SC(L_A + L_B + \frac{L_C}{2} - a)$$

$$\uparrow \sum F_y = -SA - FLG + Wing_y - Wing - SB - P - RLG - SC + Emp_y - Emp = 0$$

$$Wing_y + Emp_y = SA + FLG + Wing + SB + P + RLG + SC - Emp$$

Figure 17: In flight calculation for Structural Load Analysis

Calculations:

$$\uparrow \sum M_{Dist_{RLG}} = 0$$

$$SA(Dist_{RLG} - \frac{L_A}{2}) - FLG_y(Dist_{RLG} - Dist_{FLG}) + Wing(Dist_{RLG} - a) + P(Dist_{RLG} - Dist_P - \frac{L_P}{2}) + SB(Dist_{RLG} - L_A - \frac{L_B}{2}) - SC(L - Dist_{RLG} - \frac{L_C}{2}) - Emp(L - Dist_{RLG}) = 0$$

$$FLG_y(Dist_{RLG} - Dist_{FLG}) = SA(Dist_{RLG} - \frac{L_A}{2}) + Wing(Dist_{RLG} - a) + P(Dist_{RLG} - Dist_P - \frac{L_P}{2}) + SB(Dist_{RLG} - L_A - \frac{L_B}{2}) - SC(L - Dist_{RLG} - \frac{L_C}{2}) - Emp(L - Dist_{RLG})$$

$$\uparrow \sum F_y = 0; -SA + FLG_y - Wing - P - SB + RLG_y - SC - Emp = 0$$

$$FLG_y + RLG_y = SA + Wing + P + SB + SC + Emp$$

Figure 18: On ground calculations for Structural Load Analysis

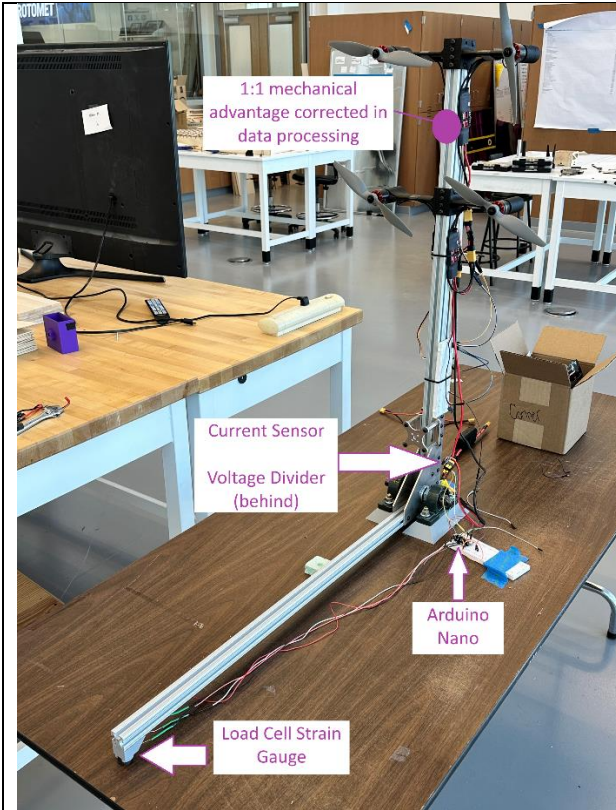


Figure 19: Labelled picture of static thrust test stand

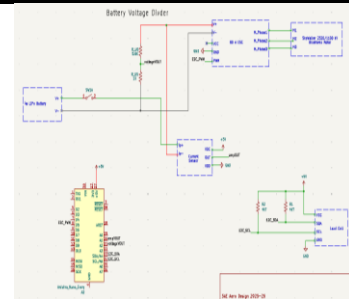


Figure 20: Circuit Diagram for Maximum Thrust Configuration of Propulsion Test Stand

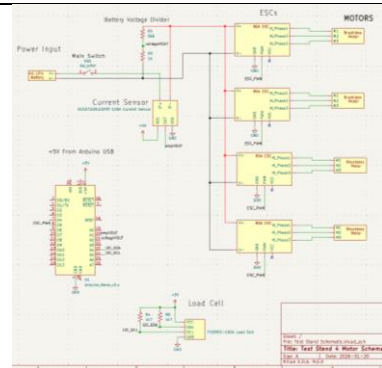


Figure 21: Circuit Diagram for endurance configuration of propulsion test stand

Appendix B (TDS)

Technical Data Sheet

Team Name: Tennessee Tech SAE Aero Design

Team Number: 034

School: Tennessee Technological University

The following discrete graph is the payload prediction generated by the process detailed in Section 6.2.6.

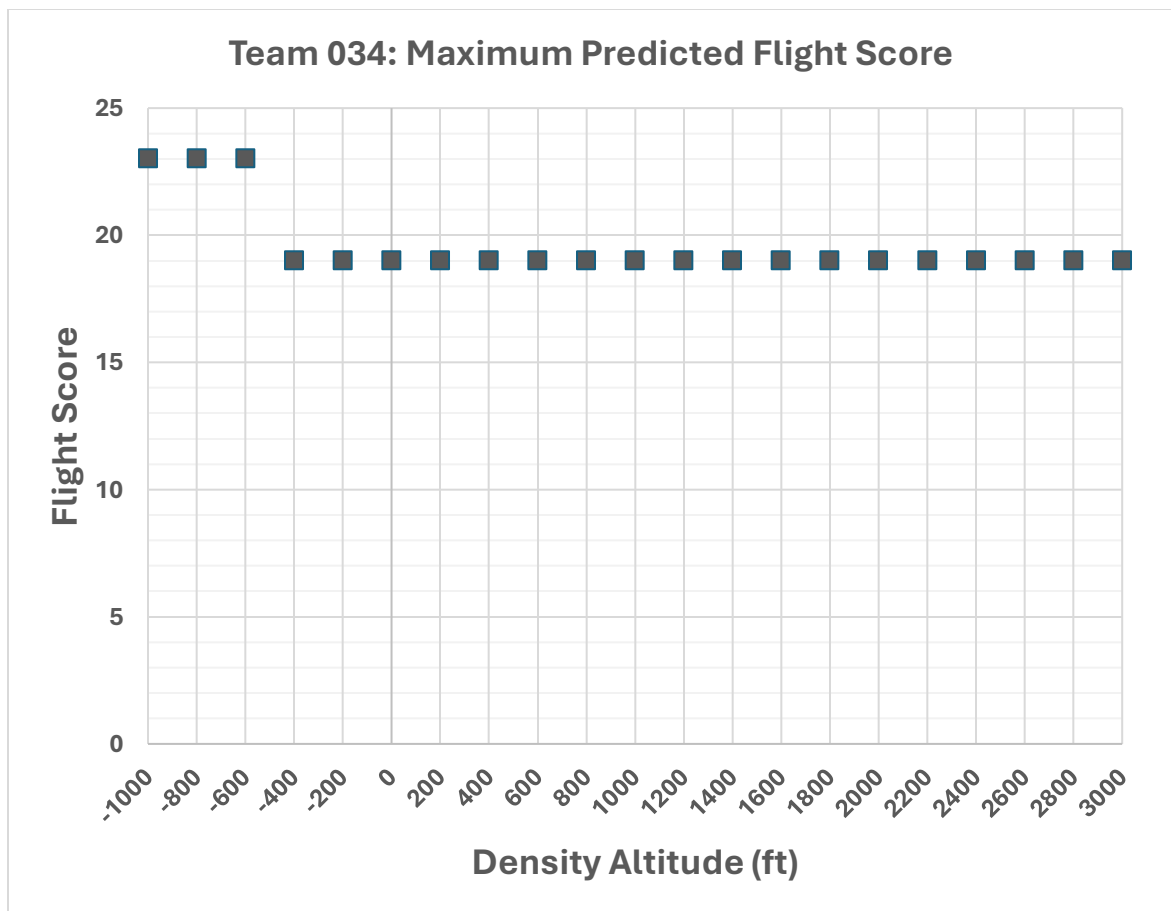


Figure 22: Payload Prediction Bonus Graph

